

2. Alloys: Mg

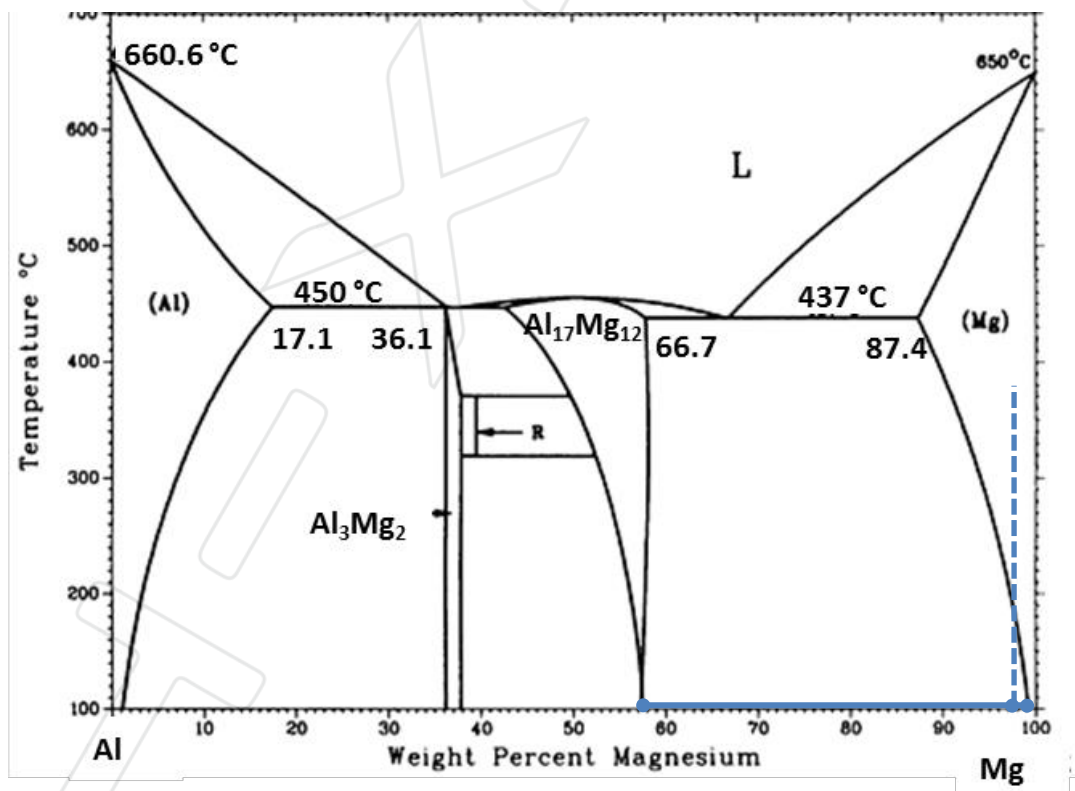
1. Magnesium (Mg)

Magnesium is the lightest structural metal on earth (density 1783 kg/m³). Its alloy can show precipitation hardening similar to Al-alloys and it can have good creep resistance up to 250 °C. However, Mg has a low melting point compared to Ti and Ni (Mg~660° C) and it has also low stiffness. It is very brittle due to the HCP structure with limited slip systems active at room temperature, so if Mg alloys need to be e.g. rolled or forged, this requires elevated temperatures. Corrosion resistance for Mg is also a concern.

2. Mg alloys

We mainly discussed two Mg alloys in our lecture.

One is AZ31 (including 3 wt% Al and 1 wt% Zn); the other is WE43 (including 4% Y, 3% Nd and Zr).



For AZ31 (Al-Mg phase diagram above), alloys are heat treated at 400° C, then after quenched, Super-Saturated Solid Solution (SSSS) will transfer directly to Mg₁₇Al₁₂ (no intermediary phases), which nucleate on (0001)Mg.

For WE43, the role of Zr is to act as nucleation sites to form fine grains even at slow cooling rate. Smaller grain size is good for (i) grain refinement strengthening and (ii) retarding crack growth. The role of Y and Nd is to have precipitation hardening, similar to Al aerospace alloys. WE43 has also good creep resistance. Numerous, small β'/β_1 ppts distributed in Mg grains are relatively stable at 250-300 °C and obstruct dislocation motion at elevated temperature. The stable ppts at grain boundaries also inhibit grain boundary sliding at elevated temperature (See more in the topic of Creep).